

1. Attempt the following.

[40 marks]

- i) $\sum_{i \leq k \leq n} O(n)$, where $O(n)$ stands for order n is
 - a) $O(n)$
 - b) $O(n^2)$
 - c) $O(n^3)$
 - d) $O(3n^2)$
- ii) _____ notation gives the tight upper bound of the given function.
 - a) Big O
 - b) Big Theta
 - c) Big Omega
 - d) Little omega
- iii) Suppose $T(n) = 2T(n/2) + n$, $T(0) = T(1) = 1$. Which one of the following is false?
 - a) $T(n) = O(n^2)$
 - b) $T(n) = \Theta(n \log n)$
 - c) $T(n) = \Omega(n^2)$
 - d) $T(n) = O(n \log n)$
- iv) A _____ is a special format for organizing and storing data.
 - a) Data structure
 - b) Algorithm
 - c) Searching
 - d) Sorting
- v) A _____ solves a problem by calling a copy of itself to work on a smaller Problem.
 - a) Method
 - b) Searching
 - c) Recursive method
 - d) Function
- vi) A _____ is an ordered list in which insertion and deletion are done at one End.
 - a) Stack
 - b) List
 - c) Queue
 - d) Linked list
- vii) An _____ is the step-by-step unambiguous instructions to solve a given problem.
 - a) Queue
 - b) Algorithm
 - c) Searching
 - d) Program
- viii) _____ is an algorithm that arranges the elements of a list in a certain order.
 - a) Searching
 - b) Sorting
 - c) Finding
 - d) Shuffling
- ix) Merge sort uses
 - a) Divide and conquer strategy
 - b) Backtracking approach
 - c) Heuristic search
 - d) Greedy approach
- x) Which of the following algorithms has lowest worst case complexity.
 - a) Bubble Sort
 - b) Quick Sort
 - c) Merge Sort
 - d) Selection Sort
- xi) _____ is rearranging pairs of elements which are out of order, until no such pairs remain.
 - a) Insertion
 - b) Exchange
 - c) Selection
 - d) Distribution
- xii) _____ order is the best possible for array sorting algorithm which sorts n item.

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- a) $O(n \log n)$ b) $O(n^2)$
 c) $O(n + \log n)$ d) $O(\log n)$
- xiii) _____ is space complexity of Binary Search.
 a) $O(1)$ b) $O(N)$
 c) $\log(n)$ d) $n \log(N)$
- xiv) _____ stores already computed values in table.
 a) Hashing b) Memoization
 c) Searching d) Sorting
- xv) Which of the following standard algorithms is not Dynamic Programming based.
 a) Bellman-Ford Algorithm for single source shortest path
 b) Floyd Warshall Algorithm for all pairs shortest paths
 c) 0-1 Knapsack problem
 d) Prim's Minimum Spanning Tree
- xvi) _____ algorithms take advantage of computer architectures to process several instructions at a time.
 a) Serial b) Parallel
 c) Complex d) Distributed
- xvii) Which of these is a postfix expression?
 a) $A + b - c$ b) $+ab$
 c) $abc * + de - +$ d) $a * b(c + d)$
- xviii) Which of the following is the disadvantage of the array?
 a) Stack and Queue data structures can be implemented through an array.
 b) Index of the first element in an array can be negative
 c) Wastage of memory if the elements inserted in an array are lesser than the allocated size
 d) Elements can be accessed sequentially.
- xix) _____ an optimal solution to a problem contains optimal solutions to sub Problems
 a) Searching b) Optimal subproblem
 c) Overlapping subproblem d) Sorting
- xx) An ADT consists of _____ parts.
 a) 1 b) 2
 c) 3 d) 4

2. Attempt the following(Solve any 02) [10 marks]

- i) What is an algorithm? Explain characteristics of algorithms.
- ii) Explain Big-O notation with proper example.
- iii) Define data structure and explain its types.
- iv) Explain stack data structure and its operations in detail.

3. Attempt the following(Solve any 02) [10 marks]

- i) Write a note on recursion. Enlist application of recursion.
- ii) Given an array: 6 8 1 4 5 3 7 2

Sort using insertion sort algorithm and state the performance.

- iii) What is searching and explain different types of linear search?
- iv) Explain Brute Force algorithm.

4. Attempt the following(Solve any 02) [10 marks]

- i) Explain different types of classification by design method.
- ii) Explain greedy strategy and different elements of greedy algorithm.
- iii) Let us consider an algorithm A which solves problems by dividing them into five subproblems of half the size, recursively solving each subproblem, and then combining the solutions in linear time. What is the complexity of this algorithm?
- iv) What is a Dynamic Programming Strategy?

5. Attempt the following(Solve any 01) [05 marks]

- i) Convert the $(X - Y / (Z + U) * V)$ infix expression into postfix expression.
- ii) Write an algorithm to find the largest element in an array A of size n and do analysis.
- iii) What is Backtracking?